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Learnovate Kit (Basic Version)

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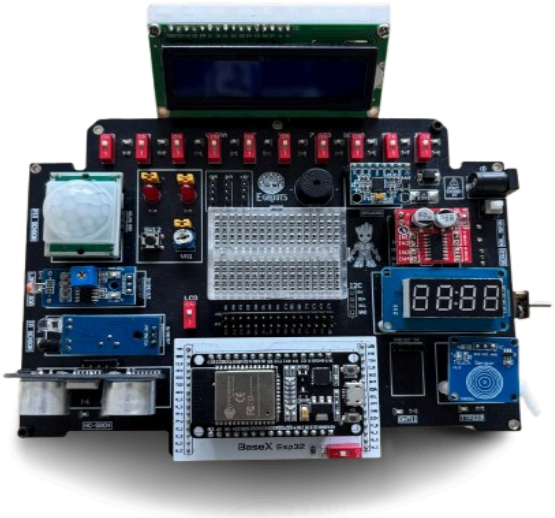
Product Overview

The **Learnovate Kit** is an educational embedded systems development kit designed to help students learn the fundamentals of electronics, microcontrollers, and hardware interfacing through practical experimentation. The kit provides a structured platform where beginners can easily understand how different electronic components and sensors interact with a microcontroller.

The kit includes a **microcontroller development board, input/output interfaces, sensor connection ports, indicator LEDs, communication interfaces, and power management circuits** integrated on a single PCB. This design allows students to perform various experiments such as LED control, sensor interfacing, motor control, and basic IoT applications.



The main objective of the Learnovate Kit is to simplify the learning process by providing a **compact, organized, and easy-to-use hardware platform**. Instead of assembling multiple modules separately, students can directly connect components and start programming, which reduces complexity and improves understanding of embedded system concepts.

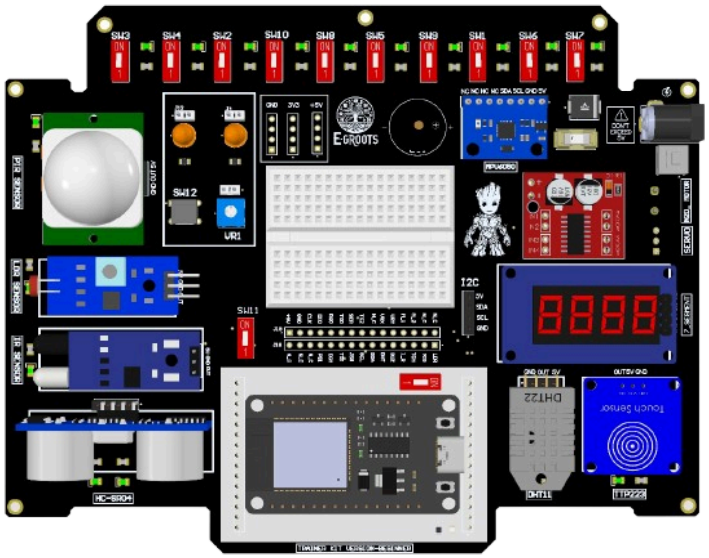
PCB Design for Learnovate Kit (Basic Version)

For the Learnovate Kit – Basic Version, my role was to design the **Printed Circuit Board (PCB)** that serves as the main hardware platform for the learning kit. The objective of the PCB design was to create a compact and user-friendly board that allows students to easily interface with different electronic components and modules for learning embedded systems and basic electronics.

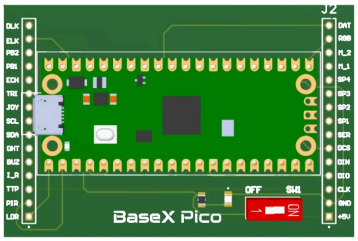
The design process began with creating the **circuit schematic**, where essential components such as the microcontroller, power supply unit, input/output headers, indicator LEDs, and communication interfaces were logically connected. This schematic ensures proper interaction between the microcontroller and the external components used in the learning kit.

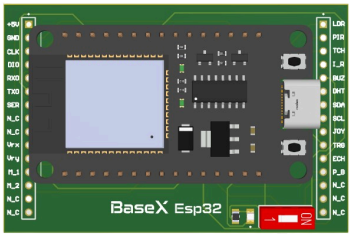
After completing the schematic, the design was converted into a **PCB layout using Altium Designer**. During the layout stage, components were arranged systematically to make the board easy to use and understand for beginners. Proper routing techniques were applied to maintain clean signal paths, stable power distribution, and minimal electrical noise.

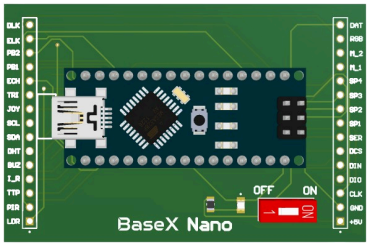
PCB Layout and 3D Design

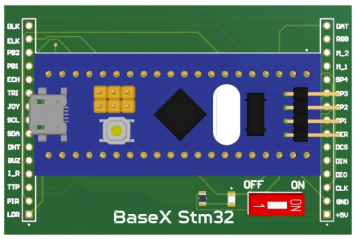
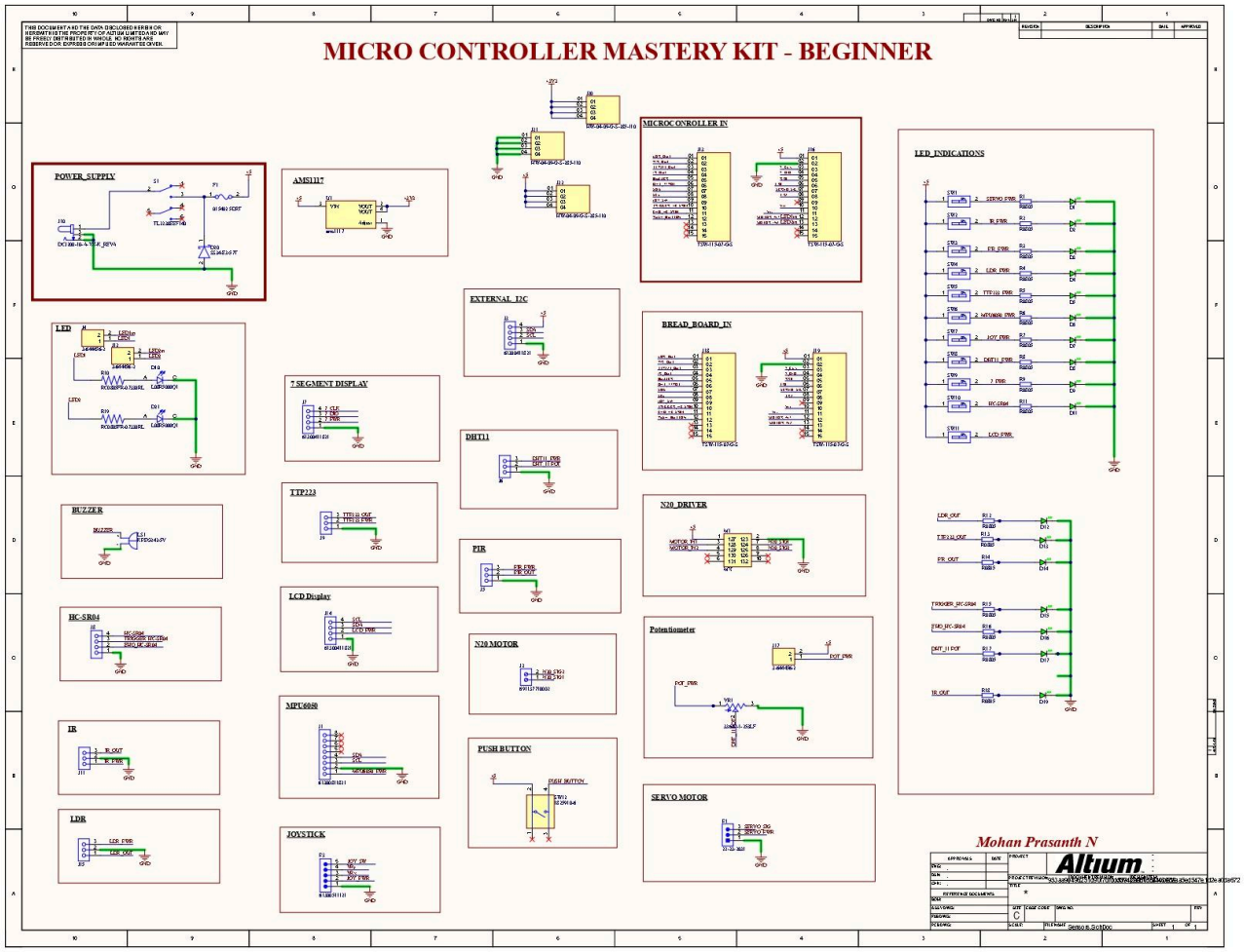


Switchable Carrier with Different Microcontrollers









Schematic Diagram